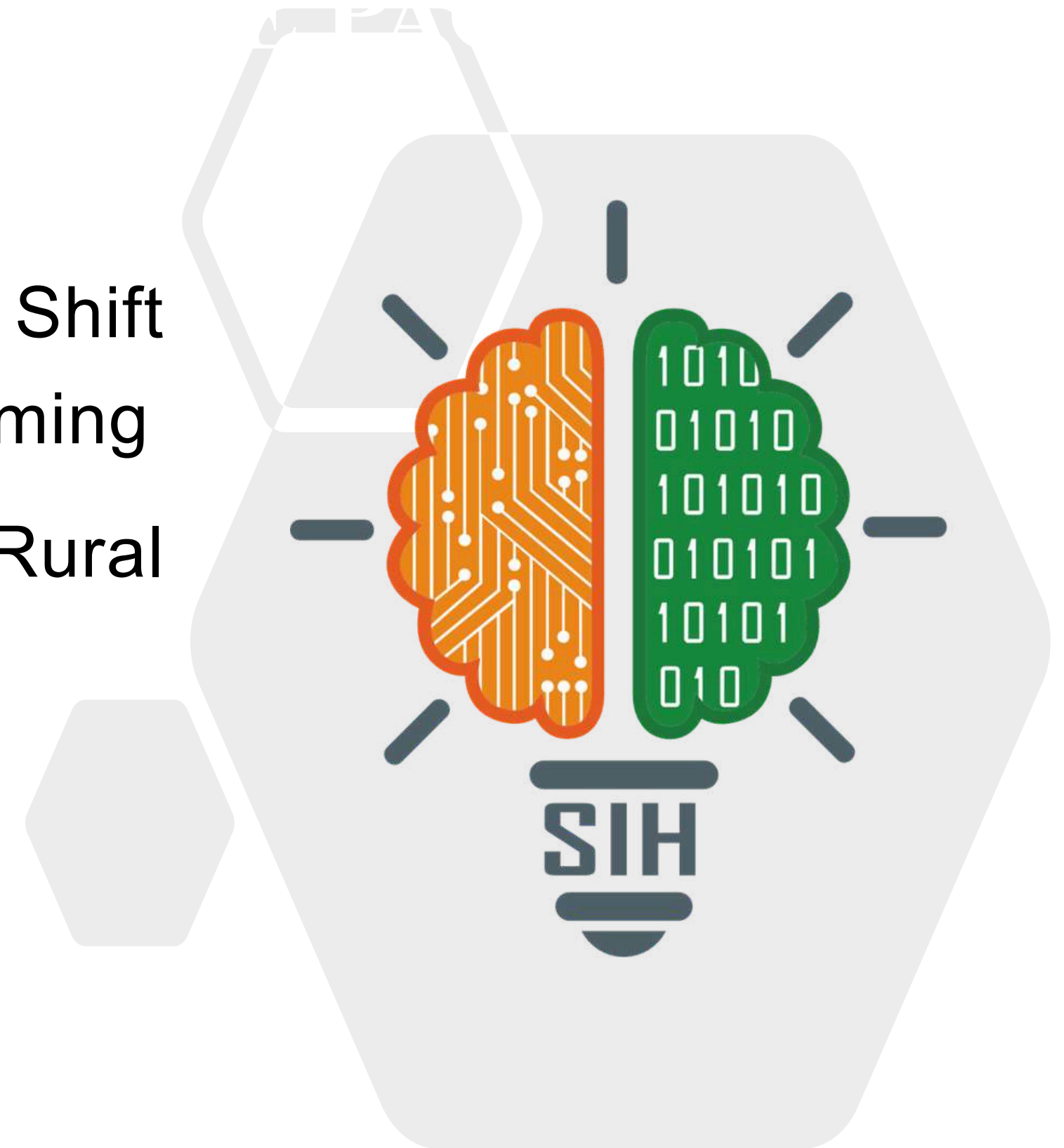


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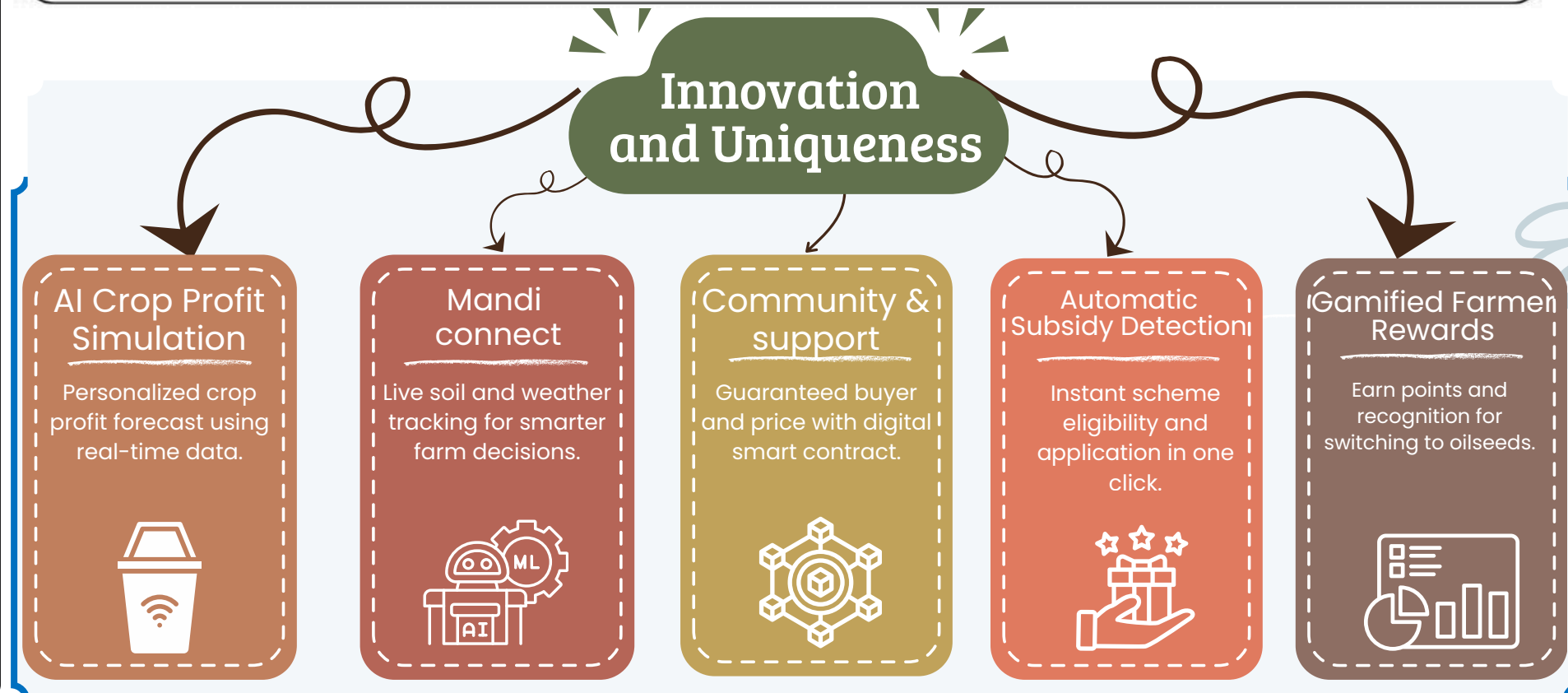
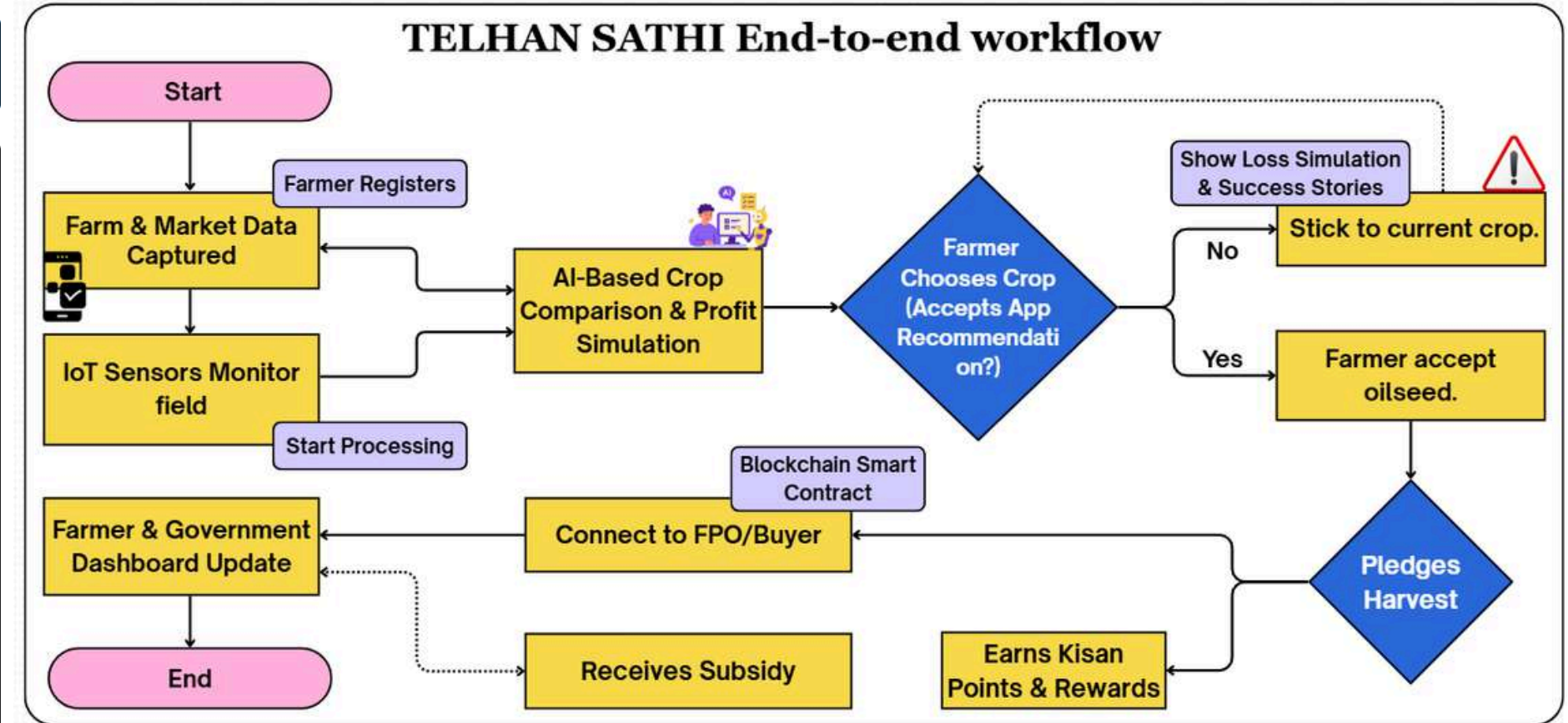
- **Problem Statement ID** – SIH25273
- **Problem Statement Title** – Crop Shift Mitigation App for Promoting Oilseed Farming
- **Theme** – Agriculture, FoodTech & Rural Development
- **PS Category** – Software
- **Team ID** – 73869
- **Team Name** – Algo Sapiens

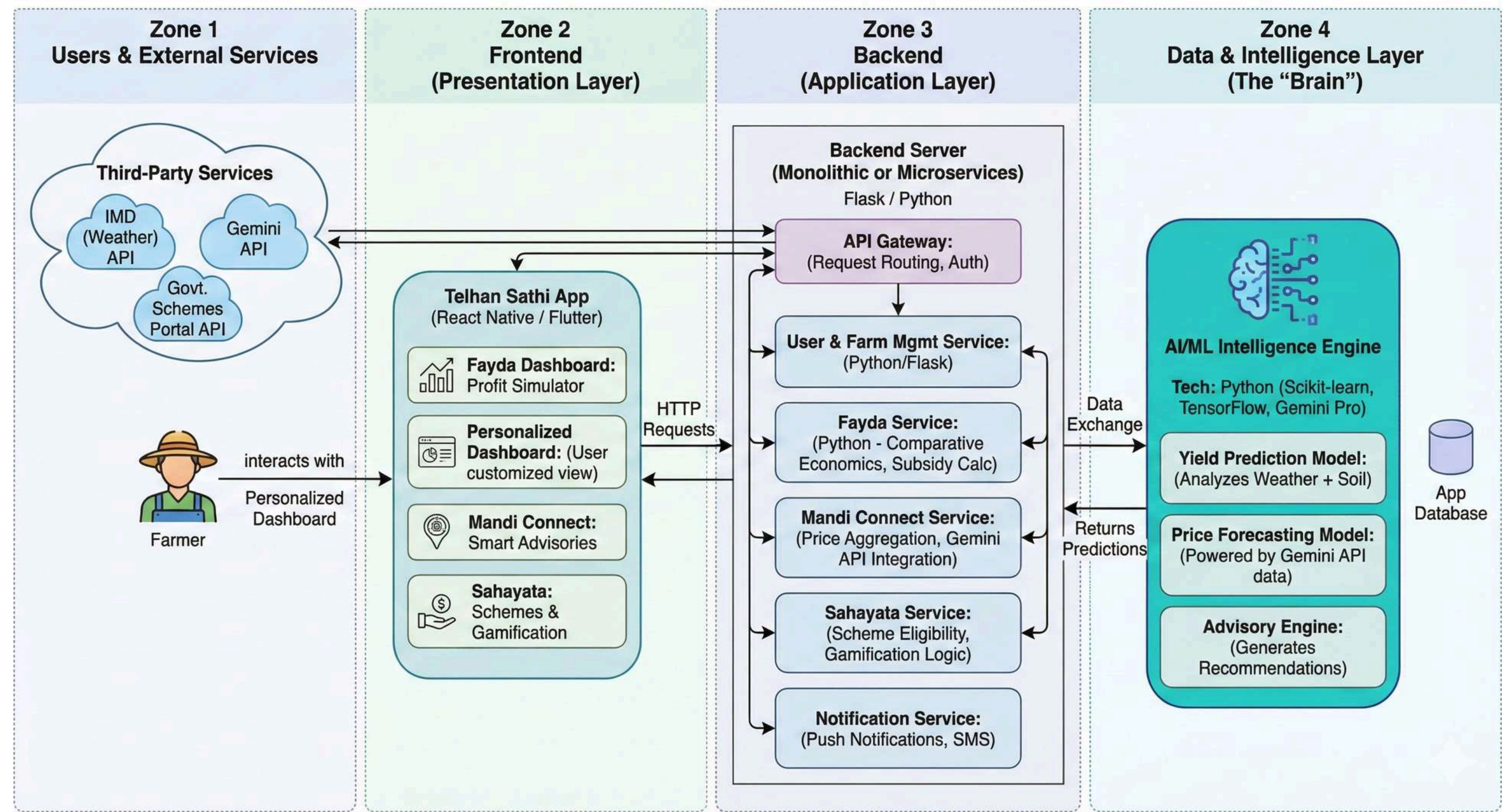




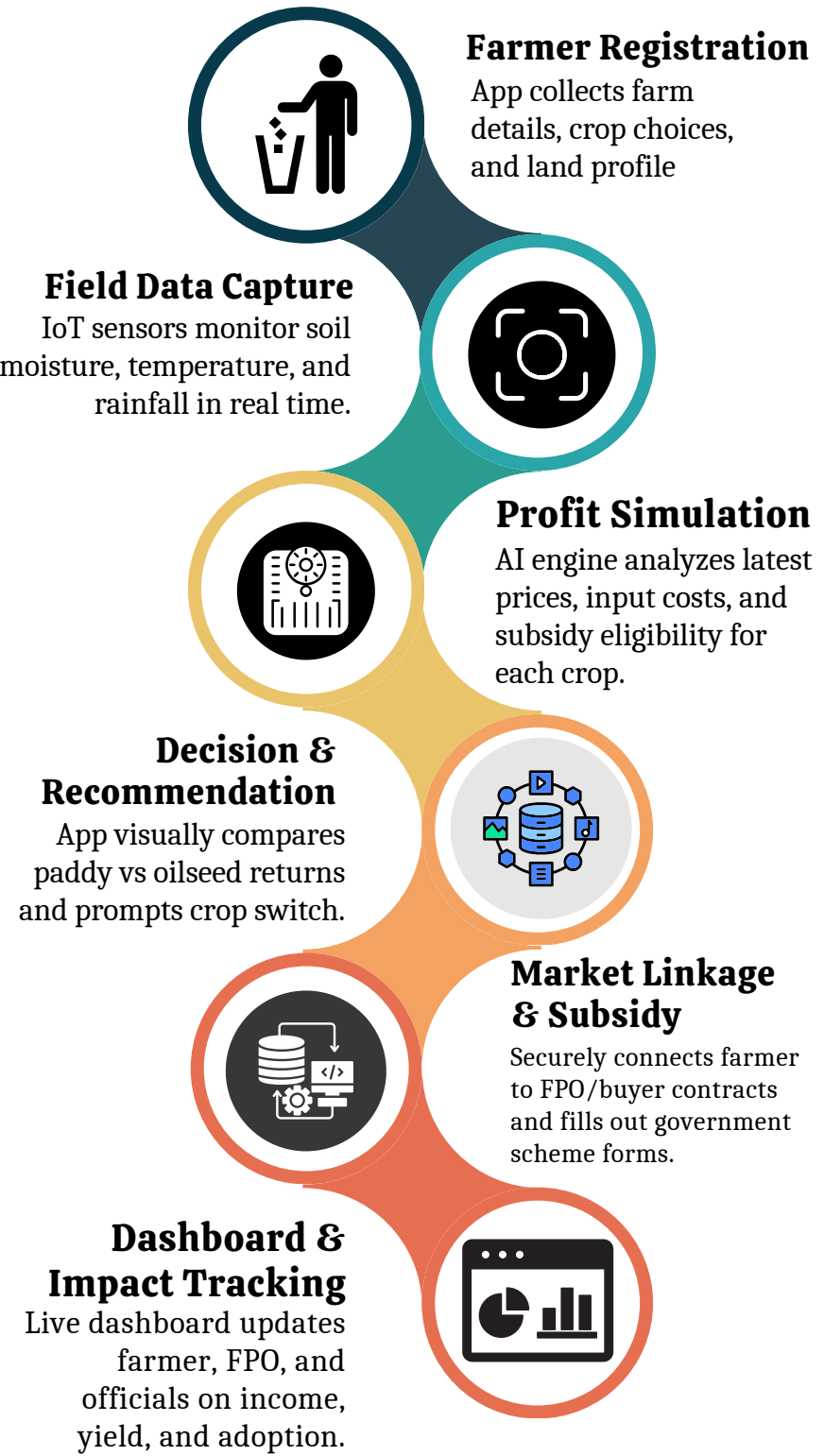
Proposed Solution / Approach

- **Assured Buyback Bridge (Mandi Connect):** Eliminates market uncertainty by creating secure digital linkages between farmers and FPOs/Buyers before the harvest season datasets.
- **Crop Economics:** Our new Crop Economics feature moves beyond simple live price display. When clicked, it shows 12 months of historical price data for that commodity/market. We present this data visually
- **Gov Support Subsidy recommender :** Automatically detects eligibility for schemes like NMEO-OS and other govt and redirect them to website
- **Fayda Forecast Engine (AI-Powered Profit Simulation)** Personalized profit forecasts for different crops and locations using real time data.
- **field nigrani :** The Field Nigrani feature integrates an IoT-based model to provide predictive analysis of field conditions. It allows for real-time monitoring and data collection to anticipate potential issues affecting crops.





Implementation Process



Components / Technology stack to be used



Feasibility

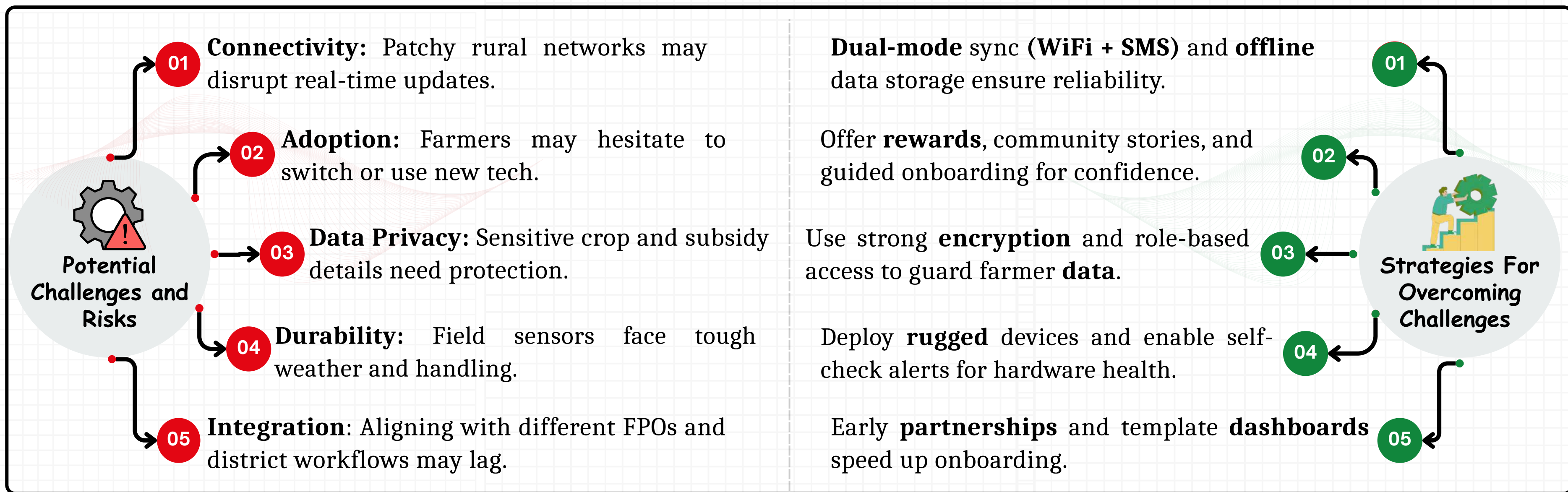
- Uses **proven** AI/ML models for accurate crop and yield prediction.
- All farm and market data is securely sent to the **cloud** for **real-time processing** and **insights**.
- **Mobile-based** OTP & **government records** ensure verified & trusted farmer participation.

Viability

- User-friendly interface and **dashboards** boost both farmer and official engagement
- Points can be claimed by farmers active participating in crop swatching can be
- **Reward** points drive active farmer Participation in crop switching and adoption.

Practical Implementation

- Ready for **pilot rollout** in real farms within **6 to 9 months**.
- Works **offline** for **remote rural areas** with seamless data sync.
- Simple **multilingual** app and UI for effortless access by all **farmers**.





Farmers: Earn **higher profits** and **reliable income** with **crop switching**, **direct buyer links**, and **smart advisories**.

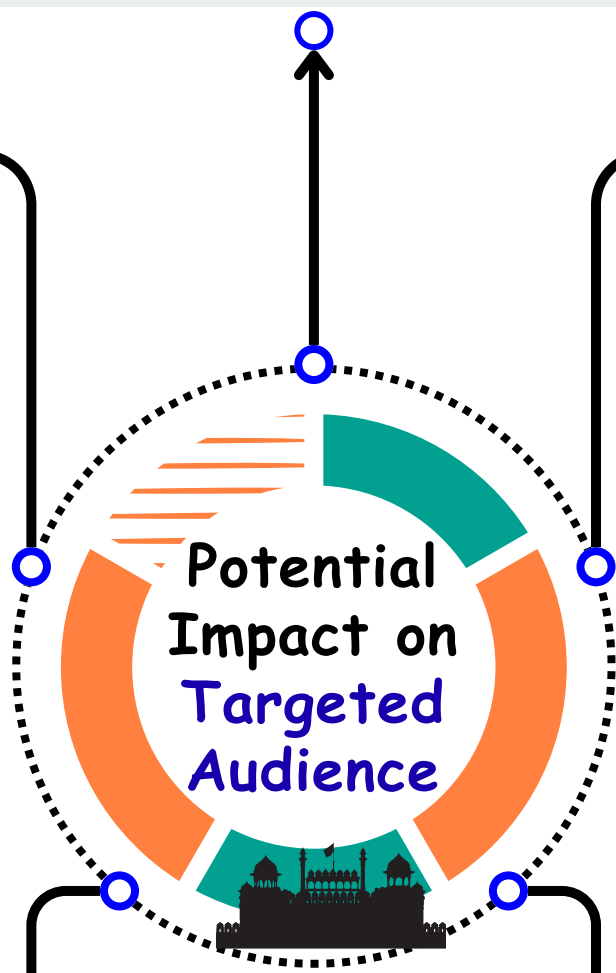


FPOs/Buyers:

Receive **quality-assured harvests** and **digital contracts**, boosting **market efficiency** and **trust**.

Nation

Reduces import dependency, strengthens **rural economies**, and supports **food security** at scale.



Government Officers:

Live **dashboards** help track **scheme adoption** and **yield growth**, simplifying **reporting** and **support**.

Community:

Peer **leaderboards** and **shared rewards** build **teamwork**, **trust**, and local pride in modern **farming**.



Benefits of the solution

“Builds **trust** with **transparent subsidy tracking**, recognized **rewards**, and peer success stories. Empowers farmer voices and increases community pride.”



Social



Economic

“Boosts **farm income** by linking to **buyers** and **subsidies**, cuts **imports**, and creates new jobs in **agri-tech** and **oilseed supply chains**.”

“Promotes **crop diversity** and **healthier soils**, reduces **unsustainable water use**, and supports **green farming** across **rainfed regions**.”



Environmental

“India’s first end-to-end traceable, community-powered waste solution—real rewards, real impact, real change.”

Field & Applied Research

- Surveyed farmer households and FPO leaders to assess crop choice drivers, mobile tech adoption, and subsidy access.
- Analyzed ICAR, Agmarknet, and e-NAM datasets for oilseed yield, price, and switching impact across diverse agro-climatic regions.

Academic & Government Sources

- **National Mission on Edible Oils—Oilseeds (NMEO-OS) Guidelines 2025:** Official Government policy, cluster schemes, and modernization targets for Indian oilseeds.. https://ganjam.odisha.gov.in/sites/default/files/2025-01/Publication%20of%20EOI%20fole%20Oil%20Oil%20Seed_0.pdf
- **ICAR-Indian Institute of Oilseeds Research (IIOR) Annual Report 2023:** Yield benchmarks, new hybrid trials, and agri-tech transfer data for oilseeds in India. https://icar-iior.org.in/sites/default/files/IIOR%20AR%202023_compressed.pdf
- **USDA Oilseeds and Products Annual (India Report 2025):** Production and yield data, market drivers, and 2025/26 growth outlook for Indian oilseed crops. https://apps.fas.usda.gov/newgainapi/api/Report/DownloadReportByFileName?fileName=Oilseeds+and+Products+Annual_New+Delhi_India_IN2025-0021.pdf
- **India Ministry of Agriculture – Production Advance Estimates Portal (2024/25):** Latest official statistics on area, yield, and production for oilseeds and other crops. <https://desagri.gov.in/en/whats-new-posts/third-advance-estimates-of-production-of-foodgrains-oilseeds-and-other-commercial-crops-for-the-2024-25/>
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